



Mahindra University Hyderabad

École Centrale School of Engineering

Minor I (2024 - Batch)

Program: B. Tech.

Branch: All

Year: II

Semester: I

Subject: Probability & Statistics (MA2103)

Date: 10/09/2025

Start Time: 10:00 AM

Time Duration: 1.5 Hours

Max. Marks: 20

Instructions:

- 1) Each question carries 5 marks.
- 2) All questions are compulsory.
- 3) Please start each answer on a separate page and ensure that you clearly number the responses.
- 4) It is essential to explain each step. Correct outcomes without any description will not be evaluated.

Q 1:

5 marks

An investor has \$20,000 to invest among 4 possible investments. Each investment must be in units of \$1000.

- (a) If the entire \$20,000 is to be invested, how many different investment strategies are possible?

20^4

[2 marks]

- (b) If the investor is allowed to invest less than \$20,000 (that is, not all the money needs to be invested), how many different investment strategies are possible?

$\sum_{n=0}^4 n^4$

[3 marks]

Q 2:

5 marks

From a city population, the probability of selecting (i) a male or a smoker is $\frac{7}{10}$, (ii) a male smoker is $\frac{2}{5}$, and (iii) a male, if a smoker is already selected, is $\frac{2}{3}$. Find the probability of selecting

- (a) a non-smoker, $\frac{2}{5}$

[2.5 marks]

- (b) a smoker, if a male is first selected.

[2.5 marks]

P.T.O.

Q 3:

5 marks

A box contains three coins with a head on each side, four coins with a tail on each side, and two fair coins. If one of these nine coins is selected at random and tossed once, what is the probability that a head will be obtained?

Q 4:

5 marks

The cumulative distribution function of a random variable X is given by

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{6}, & 0 \leq x < 1, \\ \frac{1}{3}, & 1 \leq x < 2, \\ \frac{2}{3}, & 2 \leq x < 3, \\ 1, & x \geq 3, \end{cases}$$

(a) Find the probability mass function (pmf) of X .

[2 marks]

(b) Compute $\mathbb{P}(1 \leq X \leq 2)$.

[1 marks]

? (c) Find $E[X]$ and $\text{Variance}(X)$.

[2 marks]
